

```

tb : PROCESS
BEGIN

    wait for 10 ns;
    decins <= 8;
    wait for 20 ns;

    A <= '1';
    B <= '1';
    wait for 20 ns;
    A <= '0';
    B <= '0';

    wait for 20ns;
    A <= '0';
    B <= '1';

    wait for 20ns;
    A <= '1';
    B <= '0';

    wait for 20ns;
    A <= '1';
    B <= '1';

    wait; -- will wait forever
END PROCESS;

END;
```

While creating a new test bench wave file, you require to set a few parameters on the default screen as shown in Fig. 2.

The test result for AND logic functions is shown in Fig. 3.

Program testing

First, Xilinx ISE9.2i needs to be installed in the system. In other version of the Xilinx ISE software, testing procedures vary but the code remains unaltered. The code given in this article can directly be tested by copying the 'universalboard' folder in the computer system from the EFY-CD available with this issue. Now launch Xilinx ISE9.2i from desktop on your PC and open the 'Project' option, select the 'universalboard' folder, then click 'universalboard Xilinx ISE' project. These steps will load all the relevant files into the system. The 'Project Wizard' window is shown in Fig. 4.

Steps to test the program

1. In the 'Processes' window, click the 'Create new source' option followed by

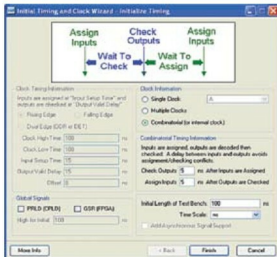


Fig. 2: Screenshot of default test bench

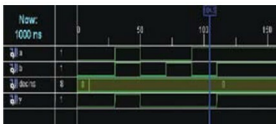


Fig. 3: Screenshot of AND logic output

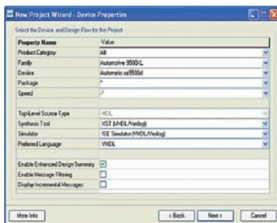


Fig. 4: Project Wizard

the 'Execute' option. Select the VHDL module and enter the file name as unilogic. Do not type any port names here; just click 'Next.' The source unilogic.vhd file is created. Now click on the 'Finish' button to complete the creation of a new source.

2. On the code page, you can type the code of the article. Entity should be unilogic in the code. After enter-

ing it, select 'Behavioural simulation' in the 'Sources' window. In the 'Processes' window, select the 'Xilinx ISE simulator' option, double click on 'Check syntax,' and then the 'Run' option.

3. Now create a test bench using 'Create New Source.' Select the VHDL test bench and enter the file name, say, 'unibw.' Note that for testing various logic functions (listed in Table IX), you have to change the corresponding decimal inputs 'decins' in the code. For example, we have used the decins value 8 for AND logic as shown in the code.

4. Select the test bench file and 'Behavioural simulation' in the 'Sources' window. In the 'Processes' window, select 'Xilinx ISE simulator' and the 'Check syntax' option.

5. When syntax is approved (indicated by a green circle), select the 'Simulate' option and run the program. The output waveform should match the truth table. The simulation output waveform, as shown in Fig. 3, exactly matches the AND truth table in Table I.

6. One can add the test bench waveform in a similar manner by selecting 'Create New Source,' 'Execute' and the 'VHDL test bench waveform' options, followed by entering the file name, say, 'unibwv.' The waveform

properties should be set at default values, unless required otherwise. Now select the Xilinx ISE simulator and run the behavioural simulation to observe the waveforms.

EFY note. All the relevant source codes of this article have been included in this month's EFY-CD. ●

Both the authors are lecturers at the Government Polytechnic Bidar, Karnataka